

QUANTT Credit Trading

Project introduction and open roles · 2026/27

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Code and full write-ups: github.com/quanttqueensu/CreditTrading

What we do

Over the summer we built a small trading desk from scratch. It runs one credit strategy on a \$500,000 Interactive Brokers paper account. The money is fake. Everything else is real: real prices, real orders sent to a real exchange, real fills, real trading costs.

The system trades by itself. Every weekday it pulls new data, checks whether it is safe to trade, sends its orders, records what came back, and raises an alarm if something went wrong. This year we want to keep it running, find out whether it survives real trading costs, and look for the next strategy.

The strategy

A closed-end fund trades on an exchange like a stock, but it issues its shares once and the count never changes afterwards. Every day it publishes what its holdings are worth, a number called the NAV. The share price is separate, and the two are often nowhere near each other.

With a normal ETF they stay close, because big banks can swap shares for the bonds inside and back again, so they trade against any gap until it closes. We watched that happen in our own data: the gap on high yield ETFs shrank from 188 basis points in 2008 to 3.8 by 2026, and that collapse killed the first two strategies we tried. A closed-end fund has none of that machinery. Nobody can open the basket, so the gap opens up and just sits there.

Our strategy compares each fund's gap to that fund's own history over the past year. Unusually cheap funds we buy, unusually expensive ones we short, equal money on each side, so it makes no difference whether the market rises or falls.

We spent much of the summer trying to prove this was an ordinary bet in disguise. Tested against high yield, investment grade, interest rates, stocks and volatility, those five explain half a percent of the returns. It also held up against the closed-end fund sector's own returns, the hardest test we could build.

Where it actually stands

The signal looks real. The trading costs might still kill it.

The strategy has to decide in the evening, because it needs the NAV and the NAV only comes out after the close. On our first live day the orders sat overnight and filled at 7:27 in the morning, two hours before the exchange opened, when almost nobody is trading these funds. We paid 0.94% to trade against a budget of about 0.10%. Priced at the numbers the strategy decided on we made \$50 that day. Priced at what we actually paid, we lost \$7,350.

We switched to orders that fill in the closing auction, the cheapest moment of the day, but we have not tested that fix because the broker platform has been down since August. Getting it back up is the first job of the year. One more thing worth saying before anyone joins: our backtest assumed an entry price

we cannot really get, and correcting it takes the expected Sharpe ratio from 0.82 down to about 0.51. We would rather recruit people on the true number.

You would not be starting from a blank folder. The summer left a daily feed of prices for 11,423 bonds built free from public filings, five scheduled jobs that run the book unattended, and a written record of twelve strategy ideas that failed and why.

Roles we are hiring

No finance background is needed. Our documentation explains the finance from scratch and assumes you know none of it.

Quant Researcher, 2 to 4 people. You test new strategy ideas and find out whether they are real. Most are not, and getting to that answer quickly is the skill. When a result looks great, your first job is to attack it. You need Python and pandas, basic statistics such as regressions and t-tests, and the temperament to drop your favourite idea when it fails a test.

Execution and Infrastructure, 1 to 2 people. You own the plumbing: the broker connection, the schedulers, the orders going out, and the measurement of what our trading really costs. This side has taught us more than any backtest, because live systems break in ways research code never shows you. You need solid Python, patience for things that fail silently, and ideally some experience with APIs or scheduled jobs.

Risk and Reporting, 1 to 2 people. You watch the live book, and each week you compare what the strategy did against what the backtest said it should do, across returns, exposures, drawdowns and costs. You need basic Python and the ability to write a short summary the team will actually read.

Goals and timeline

WHEN	WHAT
September	Recruit and onboard. Get the backtest running on every machine and the broker connection working again.
Oct – Nov	Build up to sixty live sessions, then run the review we committed to in writing before launch. If the strategy fails it, we shut it down.
Dec – Jan	Freeze the rules for the January trade, then run it. Positive in 20 of the past 23 Januaries, and it moves independently of the main strategy.
Feb – Apr	Keep researching, decide on resizing from live evidence, and write the final report and handover for next year's team.

To apply

Send a short note saying which role interests you and anything you have built, to simon.jarvis0@gmail.com. Everything is public on GitHub if you want to look first, starting with `HOW_WE_GOT_HERE.md`, the story of the summer including all the wrong turns.